



Financial Institutions & Markets- ENT 309

PPT #11, IITD, Summer, 2026

Agenda

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DERIVATIVES

- Forward contract
- Futures
- Options
- Swaps



Derivatives

- Help in risk management
- Change in factors like Interest Rates, FX rates, Commodity prices, Stock Market prices impact the value of assets
- Derivatives help in reducing the risk of loss in asset value
- Investors who want to speculate on the prices to make money can also use derivatives
- Can refer Kidwell & Blackwell for the chapter on Derivatives

Forward Contract



Forward Contract

- Traded in informal Over The Counter (OTC) market. Not on an exchange
- Between two parties. Purchasers agrees to buy an specified amount of an asset from the seller at a fixed date sometime in the future.
- Transaction to happen at the “forward price” (as against the “spot price”)
- Buyer takes a **long position**. Seller a **short position**.
- Settlement Date: Asset and payment changes hands on this date.
- Forward Contract help one cover risk to change in price.
- But exposes to Counterparty Risk



Forward Contract

- Forward Price after X days= Spot Price + Cost of borrowing an amount equal to Spot Price for X number of days.
- Which “Interest Rate” to use for calculating the cost of borrowing?
- MIBOR Mumbai Interbank Offered Rate (Refer discussion in FX classes on rates used for different currencies)
- You plan to buy, after 2 months, an existing Treasury bill which has 3-months to maturity. The total face value of securities you plan to buy is INR 5 million. The current price of the Treasury bill is INR 9,95,000 per INR 1 million face value.
- MIBOR 2-month: 6.5% per annum
- Assume simple interest. What is the forward price for delivery after 2-months?



Forward Contract

- Forward Price of the Treasury bill = ?
- Total face value of securities: INR 5 million.
- Current price for 3-month Treasury bills is INR 9,95,000 per INR 1 million face value. MIBOR 2-month: 6.5%
- Forward Price per mil Face Value = $9,95,000 \times (1 + (0.065 \times 2/12))$
= 10,05,779. (Interest compounded monthly)
- Total = $5 \times 10,05,779 = 5,028,895$
- We have done the calculation of Forward Rates in FX in our earlier lectures

Futures

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Futures Market

- Futures contracts are traded on an organized exchange. Not OTC
- Both parties in a futures market transaction hold formal contracts with the futures exchange, not with each other. Exchange is the counterparty for both seller and buyer.
- Futures contracts are standardized in quantities, delivery periods, and grades of deliverable items, whereas forward market contracts are not.
- Futures contracts call for the delivery of **specific** commodities, securities, or currencies either on **specific** future dates or over limited periods of time.
- Standardization results in a relatively large volume of transactions for a given contract. Makes trading in the contract easy and inexpensive.



Futures Market

- The pricing principle for Forward & Futures is the same: **Spot Price + Carrying cost**. In case there are any other confirmed costs or income during the period, these are adjusted.
- But the price of Futures changes daily in response to factors like spot price of asset, interest rates, time to maturity etc.
- **Margin:** Exchange runs a counterparty risk if buyer or seller default in their commitment. An amount of fund depending on the contract value is to deposited by both parties with the exchange- this is the margin.
- Since the prices fluctuate daily, the margin amount required is MARK TO MARKET. Exchange asks the participants to increase margin as required.
- The credit/debit to a participants accounts is done every EOD.
- Hence either party in the futures contract can liquidate its position by doing an opposite transaction.



Futures Market

- Example
 - If on 1st July, you promised to sell/deliver X gms of gold on 31st Aug (took a short position)
 - TODAY you can buy X gms of gold for delivery on 31st Aug (take a long position) and can make your net position zero.
- Most transactions in Futures market are offset before delivery
- In contrast, in the Forward Transaction, the asset is usually delivered on the specified date to close the transaction.
- Stop Loss facility available to limit losses.
- <https://www.nseindia.com/static/products-services/about-equity-derivatives>



Futures Market: Example

- How do Futures help in risk management? You own a bond portfolio. Interest rates are expected to rise and you are worried that you will lose money. What should you do?
- Hedge by “selling futures” : Take a short position (agree to sell at a future date) in bond futures.
- As interest rate rises, short position in futures will gain money. How?
- Bond prices fall. So futures prices also fall. You agreed **to sell** in future at X. The future price has fallen to Y now. The other party who bought futures at X loses (X-Y). You earn this amount due to mark to market margin.
- This offsets the loss incurred in your bond portfolio. Hedges the risk



Futures Market: Example

- You have a short position in a bond futures. It pays out a coupon. What will happen to the futures prices? Will you gain or lose?

The standard cost-of-carry formula is:

$$F_0 = (S_0 - I)(1 + r)^T$$

where:

- F_0 = Futures price today
- S_0 = Spot (cash) price of the bond today
- I = **Present value of all coupons expected to be paid before the futures expires**
- r = Risk-free financing rate
- T = Time to futures maturity

- So, though the price of the underlying asset (the bond) falls after the coupon payout, but this has already been incorporated in the Futures pricing. So, no impact for the short or long futures holders.

Forward Vs. Futures



DERIVATIVE

Forward Contract Versus Futures Contracts

Property	Forward Contract	Futures Contract
Nature	Private contract between 2 parties	Standardized contract
Trading Venue	Over the counter (OTC) transaction	Organized exchange
Terms	Customized & flexible terms	Standard quantity, quality, maturity
Risk	Counterparty (CP) Risk	Nil CP risk. Exchange as mediator
Liquidity	Normally low	Very high
Settlement	At maturity	Sell anytime at Mark to Market
Price Information	Private	Publicly available
Delivery	Mostly delivery of FX happens	Most contracts closed before delivery

Forward Vs. Futures



DERIVATIVE

Forward Contract Versus Futures Contracts

- Suppose an Indian importer needs exactly USD 1,10,000 after 62 days.
 - A bank creates a specific forward contract
 - Amount: USD 1,10,000
 - Maturity: 62 days
 - Exchange rate: agreed today
- Futures Contract: A futures exchange may offer:
 - Contract size: USD 100,000
 - Expiry: March, June, September, December
 - The importer must choose among available contracts.
- Typical Users
 - Forward Contracts – Exporters, Importers, Banks, Corporations
 - Futures Contracts – Hedgers , Speculators, Arbitrageurs, Institutional investors.

Options Market



Options Market

- Futures have a drawback. While you are safe from losses, you cannot make any potential profits also.
- Options allow people to enter into contracts to buy or sell stocks, commodities, or other securities at a predetermined price, called **the strike or exercise price**, until some future time.
- An option buyer **has the right** to buy or sell but **no obligation** to do so.
- The person who sells the option has to execute the transaction to buy or sell the underlying asset if the option is “exercised” by the option buyer.
- As compensation, the option buyer pays the option seller a premium. This is the price of the option.
- Option prices can be calculated using **Black-Sholes model**. Myron Sholes was a founding member of **LTCM**



Options Market

- Option is a one-sided choice. If it is beneficial to exercise the option, the option buyer will do so. Else he will allow it to expire and just lose the premium.
- **Call Option:** Give the buyer the right (but not obligation) **to buy** a security or futures contract at the strike price.
- **Put Option:** Buyer of the option has the right but not an obligation **to sell**.
- Gain and loss of the Option buyer and Option seller is limited by the premium paid for the option.
- Comparatively, in futures, the loss and gain are **not capped/ are unlimited** but are symmetrical.



Options Market: Example

- You own a bond portfolio. Interest rates are expected to rise and you are worried that you will lose money. **What should you do?**
- If you go short futures, your losses or gains will be hedged.
- You can buy a put option. If interest rates rise, the value of put option will go up (You have purchased an option to sell the bonds at price X, but now the bond prices are lower. So the “put” is more valuable). Exercise the put option and benefit from the hedging.
- If interest rates fall, your bond portfolio gains. So you can let the put option expire. Only expense is the price of the 'put option' that you have paid.

Swaps

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FX Market Trxns



DERIVATIVE

FX SWAP TRANSACTIONS

- Do opposite transactions (Buy/Sell or Sell/Buy) simultaneously.
- Combination of a Spot txn done today & a Forward txn done for a future date
- Example: You import crude oil from US and sell them petroleum products after 3 months. Sell INR & Buy USD at Spot Rate today to pay for imports. Contract to sell USD after 3 months at 3 month Forward rate today.
- Hedge against exposure to USD fluctuations.
- Forward Contract: When FX involved only on one leg of the txn
- FX Swap Transaction: When FX involved on both legs
- *Futures & Options are also available as FX derivatives*



Thank You!